Department of Chemistry
Jahangirnagar University
1st Tutorial Examination
Chem-315, Marks: 20, Duration: 1 hour

1. Sketch a graph to show the depression of freezing point of a solvent due to addition of a solute. Also, write the mathematical expression for determining the molecular weight of a solute using the concept of cryoscopic method. 4
2. Mention the differences between volumetric and conductometric titration. 2
3. (i) Write the mathematical formula of Beer-Lambert Law. What are the limitations of Beer-Lambert Law? 4
   (ii) Give the absorption curve of $KMnO_4$ and locate the maximum absorption.
4. How could you construct a standard curve for determining the concentration of an unknown solution? 3
5. Discuss the potentiometric titration of HCl acid solution using NaOH solution giving neutralization curve. 3
6. Define the terms: equivalent conductivity at infinite dilution, partial molar volume and specific rotation. 4

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Tutorial Examination

)Course – 315: Time; 45 min

Marks - 15

Q.1. a) Define colligative properties. Write equation showing relation between molecular mass of a non-volatile and non-electrolyte solute and depression of freezing point 1.5+1

b) When ΔT_f will be equal to K_f? 1

c) What is the freezing point of a solution containing 3.42 g of sucrose and 500 g of water. (M.W suc.- 342, $K_f = 1.86$) 2

d) Will depression of freezing point of 1 mole of sucrose and 1 mole of NaCl be equal? Give reason. 2

Q.2. a) Define adsorption and adsorption. 4

b) Write equation for Freundlich and Langmuir adsorption isotherm. Show the validity of the law. How could you calculate slope and intercept from the plot? values of slope and intercept of them from straight line equation.

Q.3. a) Draw the phase diagram of three component system and show the homogeneous and heterogeneous area in the diadram.

Q.4. a) Define partial molal volume. 2

b) Calculate the value of V_m for the mixture of A na B having the following parameters- Mole fraction of A =0.1; MW of A=60; MW of B=80; Density of the mixture = 0.957 g/cm^3.

Q.5. a) Draw plots showing validity of Beer – Lambert. Write the equation of straight line for the plot. How could you calculate unknown concentration from the plot?

Tu-12(a)

2nd Tutorial Examination

Tu-12(b)

)Course - 315: Time: 45 min

Marks - 15

Q.1. a) Write the reaction showing hydrolysis of the ester you have use in the experiment. 1

b) Describe how could you determine rate constant from conductance measurement? 4

c) Write the unit of rate constant for second order reaction. ($L mol^{-1} s^{-1}$) 1

Q.2. Draw and explain titration curve for the titration of weak acid by a strong base. 4

Q.3. What is standard electrode potential? 2

Q.3 An electrochemical cell is constructed from a silver half-cell and a copper half-cell. The copper half-cell contains 0.10 M $Cu(NO_3)_2$ and the concentration of silver ion concentration is 2.5×10^{-7} M. If the Ag electrode is the cathode,

a) write the cell notation. 1

b) write the oxidation and reduction half reaction. 2

c) Which electrode will be positive? 1

d) calculate the emf of the cell. 4

(i) Conductance

(ii) molar conductance

(iii) Rate constant ($CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$)

1st Tutorial Examination
Course – 315; Time; 45 min
Marks - 20

Q.1. a) Write equation showing relation between molecular mass of a non-volatile and non- electrolyte solute and depression of freezing point. Will electrolyte and non- electrolyte depress freezing point equally? Explain the reason. 4

b) When ΔT_f will be equal to K_f? 1

Q.2. a) Define adsorption isotherm. 1.5

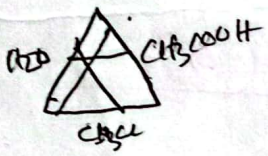
b) What is activated charcoal? Why charcoal in powder form adsorb more than same amount of solid? Draw plots showing validity of Freundlich and Langmuir adsorption isotherm. Give the values of slope and intercept of them from straight line equation. 1+1.5+3

Q.3. a) Calculate the degree of freedom for three component system. 1.5

b) Draw the phase diagram of three component system and show the homogeneous and heterogeneous area in the diadram. 1.5

Q.4 a) Calculate the value of V_m for the mixture of A na B having the following parameters- Mole fraction of A =0.1; MW of A= 60; MW of B=80; Density of the mixture = 0.957 g/cm^3. 2

Q.5. a) Draw plots showing validity of Beer –Lambert. Write the equation of straight line for the Plot. How could you calculate unknown concentration from the plot? 3



2nd Tutorial Examination
Course - 315. Time; 45 min
Marks - 20

Q.1. a) What is hydrolysis? Explain by drawing graph how could you determine the rate constant from a base catalyze hydrolysis of an ester. 5

b) Write the unit of rate constant for second order reaction. 1

c) Define molar conductivity. Give unit of molar conductivity. 1.5

d) Why equivalent conductance increases with dilution? 1.5

Q.2. Draw and explain titration curve of weak and strong acid by a strong base. 6

Q.3 What is salt bridge? Why it is used in electrochemical cell? 2

Q.4. Calculate the potential of the following cell at 25^{0}C 3

$$Zn(s)\,|\,Zn^{2+}(0.020\ M)\,||\,Cu^{2+}(0.020\ M)\,|\,Cu(s)$$

$Cu^{2+} + 2e \rightleftharpoons Cu(s)$; $E^0 = 0.337$ V

$Ag^+ + e \rightleftharpoons Ag(s)$; $E^0 = 0.799$ V

$Cu/Cu^{2+}\,||\,Ag^+/Ag$

anode oxi. cathode red.

$E_{cell} = E_{anode} + E_{cathode}$ $+0.799$ V $=0.$

1st Tutorial Examination

Course -- 315; Time; 45 min

Marks - 15

Q.1 a) Why freezing point depression occurs when a non volatile and non electrolyte solute is added to a solution? 1

b) When ΔT_f will be equal to K_f? 1

c) Name colligative properties. Why colligative properties of electrolytes and non electrolytes are not same? 1+2

Q.2. a) What is activated charcoal?

b) Define adsorption isotherm. Write Langmuir and Freundlich adsorption isotherm equation and show their validity by drawing plots. 2+1+2

Q.3. a) Calculate the degree of freedom for three component system. 1.5

b) Draw the phase diagram of three component system and show the homogeneous and heterogeneous area in the diadiam. 2.5

Q.4. a) Define partial molal volume. 2

b) Calculate the value of V_m for the mixture of A and B having the following parameters- Mole fraction of A =0.1; MW of A= 60; MW of B=80; Density of the mixture = 0.957 g/cm 2

Q.5. a) Draw plots showing validity of Beer –Lambert. Write the equation of straight line for the Plot. How could you calculate unknown concentration from the plot? 2



1st Tutorial Examination

Course – 315; Time; 45 min

Marks - 10

Q.1. a) Define colligative properties. Write the name of the colligative properties. 1

b) Depression of freezing point of 1 mole of sucrose and 1 mole of NaCl will not be the same. Give reason. 1.5

Q.2. a) What is activated charcoal? 0.5

b) Write Langmuir and Freundlich adsorption isotherm equation and show their validity by drawing plots. 2

Q.3. a) Define partial molal volume. 0.5

b) Calculate the value of V_m for the mixture of A na B having the following parameters: 2

Mole fraction of A =0.1; MW of A= 60; MW of B=80; Density of the mixture = 0.957g/cm^3.

Q.4. a) Draw plots showing validity of Beer –Lambert law. Write the equation of straight line for the plot. How could you calculate unknown concentration from the plot? 2.5

1st Tutorial Examination

Course – 315; Time; 40 min

Marks - 10

Q.1. a) Define colligative properties. 1

b) What is the freezing point of the 0.33 *m* solution of a nonvolatile nonelectrolyte solute in benzene. (fp of Benzene = 5.5 °C, K_f for benzene = 5.12 Cm^{-1}) 2

Q.2. a) Define adsorption and adsorption. 2

b) Write equation for Freundlich and Langmuir adsorption isotherm. Show the validity of the law graphically. How could you calculate slope and intercept from the plot? 2

Q.3. a) Define partial molal volume. 1

b) Calculate the value of V_m for the mixture of A and B having the following parameters: Mole fraction of A =0.1; MW of A= 60; MW of B=80; Density of the mixture = 0.957 g/cm^3. 2

2nd Tutorial Examination

Course - 315; Time; 40 min

Marks - 10

Q.1. a) Draw plots showing validity of Beer –Lambert. Write the equation of straight line for the Plot. How could you calculate unknown concentration from the plot? 2

Q.2. a) Show hydrolysis of the ester you have use in the experiment. 1

b) Molar conductivity of a weak electrolyte increases with dilution- explain why. 1

Q.3. Draw and explain titration curve for the titration of weak acid by a strong base. 3

Q.4. a)Write cell notation of the cells prepared by Zn and Cu with saturated calomel electrode In your experiment. 1

b) Show graphically and from equation how could you calculate standard electrode potential of E^0 of Zn and Cu. 2

$Zn/Zn^{2+} \,||\, Cu^{2+}/Cu$

Department of Chemistry, JU
1st Tutorial Examination 2021
Course # Chem-315, Full Marks: 20, Time: 1 hr.

1. Define the terms: 2
(i) Equivalent conductivity at infinite dilution and (ii) Specific rotation.
2. Describe the principle of determining molecular weight of a substance by cryoscopic method. 3.5
3. Write the mathematical formula of Beer-Lambert's law and give the significance of each symbol involved in the equation. How could you investigate the validity of this law? 1+2=3
4. Discuss the conductometric titration of weak acid with strong base giving neutralization curve. 4
5. Sketch the neutralization curves for the potentiometric titration of strong and weak acids with NaOH. Also draw differential curve of $\Delta pH/\Delta V$ versus volume of added alkali and locate the end point of titration. 2+2
6. Write the mathematical form of Langmuir isotherm. Also sketch the plot of Langmuir isotherm. How could you proof the validity of this isotherm? 3.5